AI infrastructure for pollinator resilience

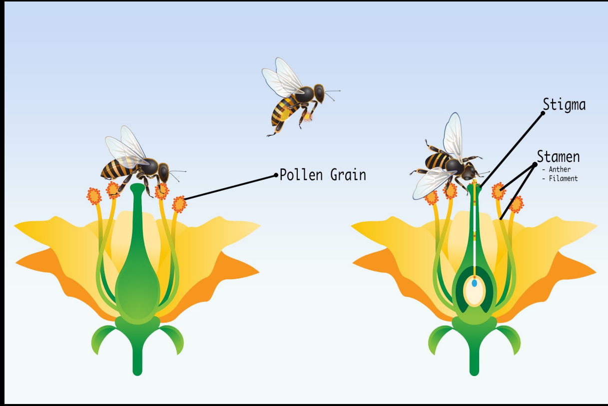
helping semi-professional beekeepers detect colony risks earlier,
reduce avoidable losses, and scale healthy pollination capacity.

Artjom Kurapov
Founding engineer

Problem statement - Food security

Bees are critical to pollination and food-system resilience

- World population to reach 10 Billion, limited resources, need of advanced food production
- 75% of leading crop types are dependent on bees pollinating them (coffee, tomatoes, apples, almonds)
- Farmers can increase crop yields by +37% with precise pollination
- **Beekeepers** providing services to **farmers** earn 9x more than from selling honey
- Demand of pollination grows 2x faster than growth of honeybee colonies






Problem statement - Efficiency

- Beekeepers lose 20-50% of colonies every year, a single colony loss impact > 350 EUR
- Bees swarm, get infested with mites or can be aggressive
- Beekeepers need to perform weekly inspections.
- Common beehives are 150 years old and heavy to inspect
- Physical labour is hard to scale, it is a seasonal activity
- Lots of beekeepers develop and use custom monitoring solutions



Vision

app

Data analytics app for beekeepers

Model beehives you own

Accepts manual upload of frame photos

Performs AI detections (bees, queen, cells)

Provides AI advices based on aggregate detections

Provides API for custom hardware

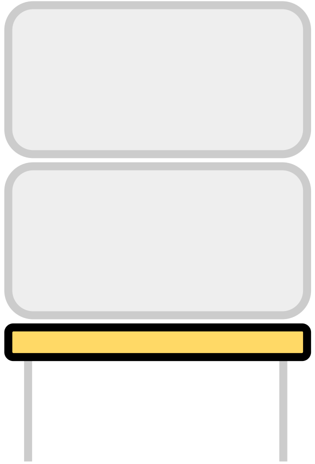
Vision

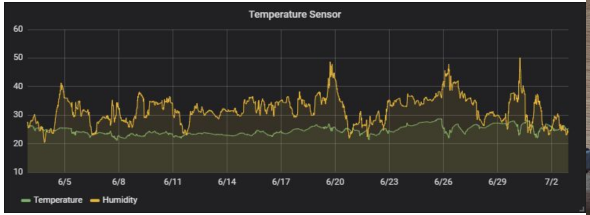
app

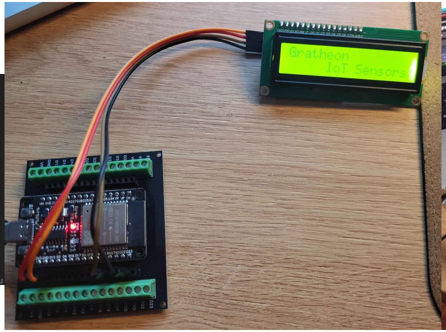
Beehive IoT sensors

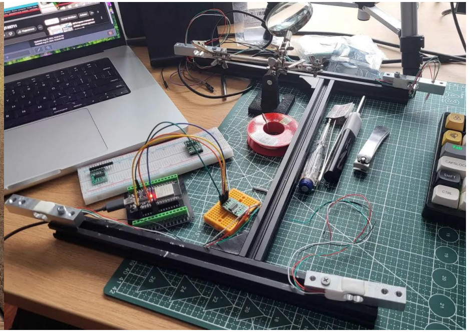
Affordable set of scales and sensors as beehive base

Sends hive internal temperature, weight, humidity
AI detects anomalies
Sends alerts in case of swarming, storms, bear attack



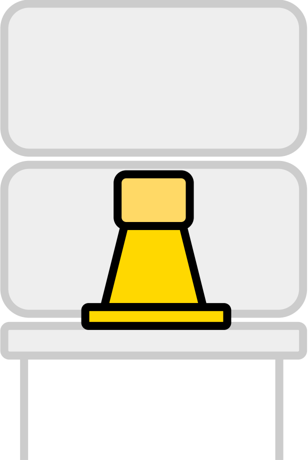






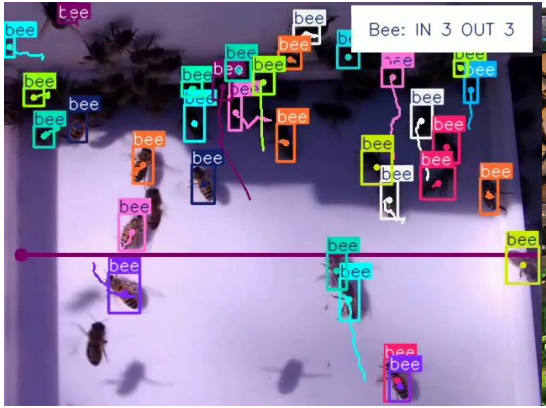
Vision

app → Beehive IoT sensors → Entrance Observer



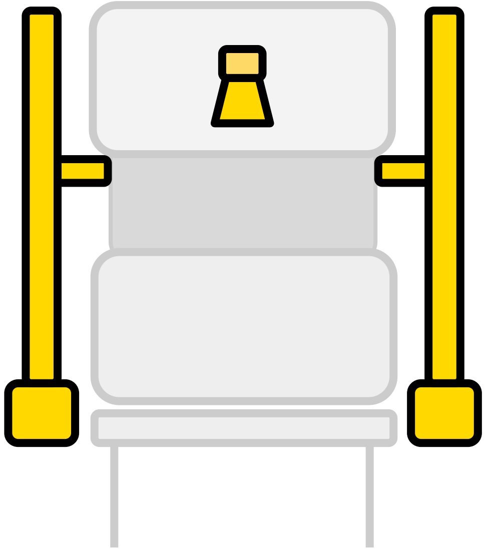
Hive entrance video monitoring device

Incoming/Outgoing bee count to estimate colony strength
Hornet and Varroa mite detection
Video streaming & playback



Vision

Beehive section lifting mechanism
Camera to capture photos of the hive frames
On wheels to manage multiple hives

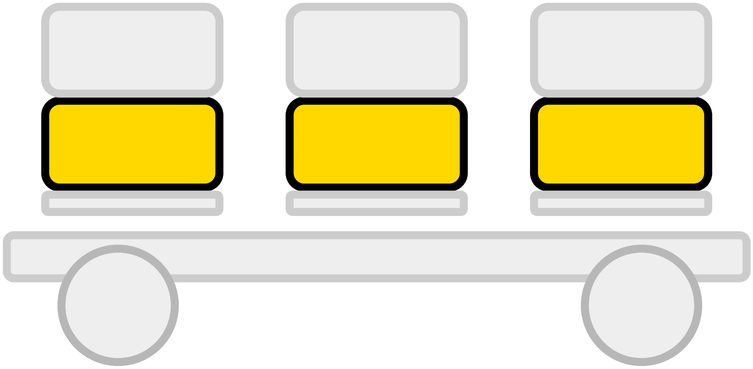


Vision

app → Beehive IoT sensors → Entrance Observer → Robot → Robotic Apiary

Automatic inspections of multiple hives

Car-towed / mobile for easy positioning in the field



Customer

Addressable market - **370 thousand semi-professional beekeepers in Europe**

Europe in total has 620k beekeepers, 19-25M colonies

~ 60% beekeepers have > 25 bee colonies

> 50% have legal company (thus B2B)

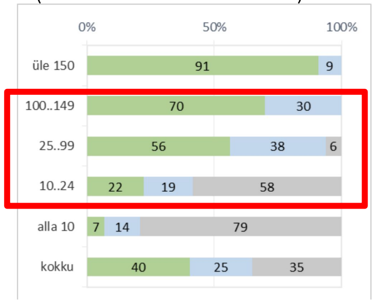
Table 1 – Number of beekeepers in selected EU countries

EU countries with more than 20 000 beekeepers	Total number of beekeepers	Beekeepers with >150 hives	
		Number	Average No of hives
Germany	116 000	81	587
Poland	62 575	324	272
Italy	50 000	2 000	413
Czech Republic	49 486	107	260
France	41 560	1 717	366
United Kingdom	37 888	50	443
Austria	25 277	380	233
Greece	24 582	7 288	165
Spain	23 816	5 361	406
Romania	22 930	1 545	194
Hungary	21 565	1 546	218

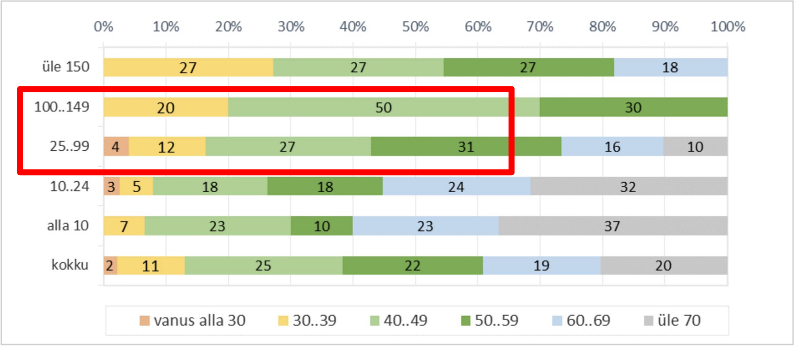
Additional users - hobby beekeepers

Early adopters - young, tech-savvy beekeepers

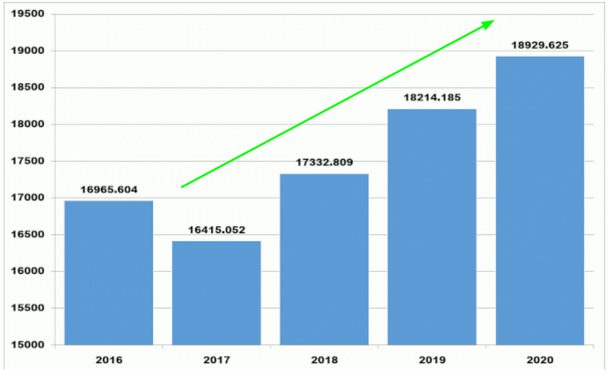
percentage of customers registered as company, depending on amount of hives, (based on estonian market research)



age distribution depending on hive count (based on estonian market research)



number of beehives in EU over the years (in thousands)

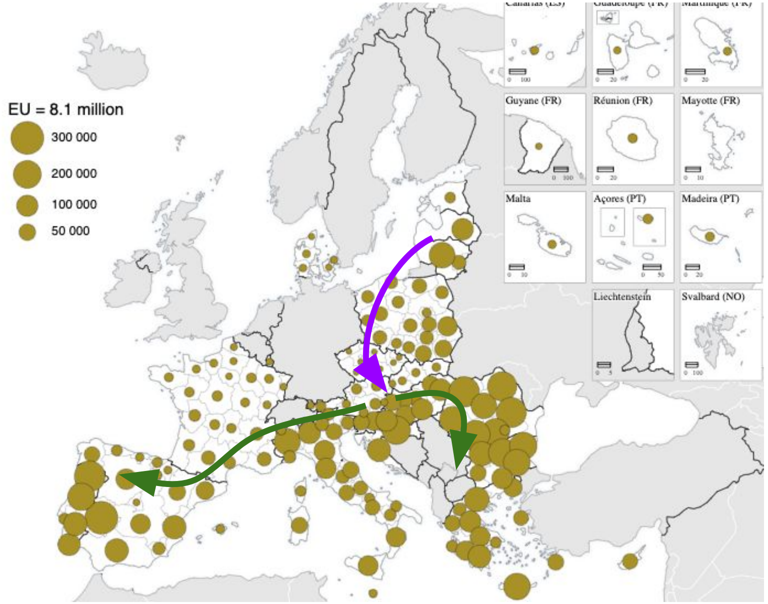


💶 Business model

Subscription model for data management and analytics, usage dependent

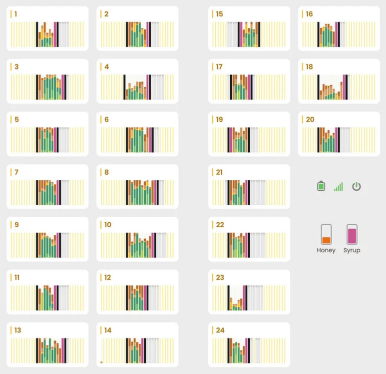
Low-margin hardware with open hardware and software to ease adoption and trust

Moat - hardware-to-software integration,
Hard to migrate (telemetry) data out



🏇 [Competition](#) in AI vision and robotics

- [beewise.ag](#) - robotic multi-colony container hive, total raised 120M $
- [beehero.io](#) - IoT, total raised 64M $
- [nectar.buzz](#) - SaaS, raised 820k $
- [beemate.buzz](#) - counts bees
- [apic.ai](#)
- [bestbees.com](#)



Traction

- product already built across app + alerts + video infrastructure
- 200+ registered users
- active open-source contributor base
- early validation from researchers and beekeeper community
- field-testing pipeline forming

Team

Research and engineering heavy team
with unique company values

Research advisors, Estonia

Artjom Kurapov
Founding fullstack engineer,
beekeeper
(ex-Pipedrive, Clarifai)

Aleksei Prokopov
Robotics, backend engineer
(ex-Fits.me, ex-Coop)

Vjatšeslav Kekšin
Researcher, PhD student
TalTech

🍯 Invest

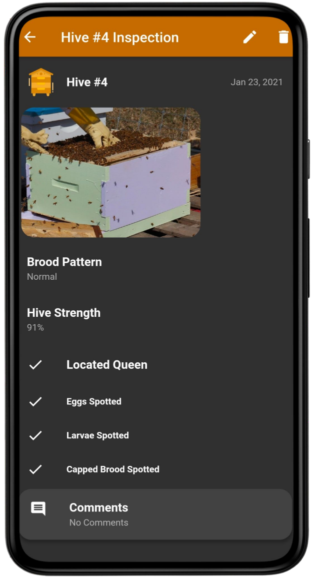
We are pre-seed and looking for **mission-aligned investors** to accelerate field validation, commercial pilots, and productization.We are pre-seed and looking for mission-aligned investors to accelerate field validation, commercial pilots, and productization.

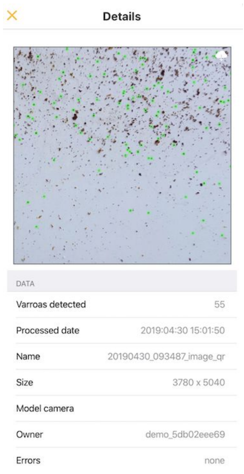
- Min. 2 summers are needed for field testing
- **Team of 4** + external contractors & beekeepers
- IoT sensors product development and release to the market
- Field testing with local beekeepers
- Entrance observer product development
- Robot prototype development

pilot@gratheon.com

Competition - Data organizer apps

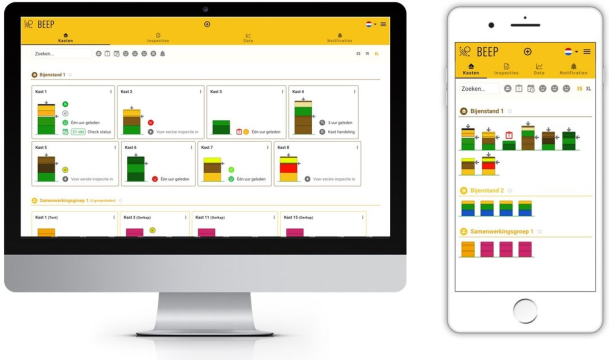
- BeeScanning
- ApiZoom
- HiveTracks
- HiveBloom
- BeeQueenDetector
- apimanager
- apiary book





🏇 Competition - IoT (audio, humidity, temperature)

- 3bee.com
- beep.nl - opensource
- broodminder.com
- beelab.se
- intelligenthives.eu
- beehivemonitoring.com
- solutionbee.com
- beehivemonitoringusa.com
- osbeehives.com
- beesage.co



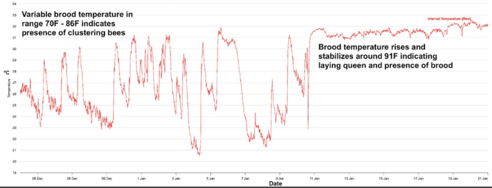


Fig. 2: Using Brood temperature to detect onset of laying queen in late winter/early spring